# AutoAntibiotic: A Reproducible Virtual Screening Pipeline for MRSA PBP2a Reveals the Insufficiency of Rigid Docking for Shallow, Flexible Active Sites

July 30, 2026


**Abstract**

Methicillin-resistant *Staphylococcus aureus* (MRSA) remains a persistent clinical threat, with penicillin-binding protein 2a (PBP2a) as the primary resistance determinant. We present the AutoAntibiotic Discovery Pipeline v5.6.0, a fully reproducible computational framework for structure-based virtual screening against MRSA PBP2a. The pipeline performs multi-conformer ensemble docking across three PBP2a conformers (PDB 3ZG0, 1VQQ, 4DKI) using AutoDock Vina 1.2.7, with induced-fit docking (IFD), PROPKA-based protonation assignment, and multi-replica explicit-solvent MD validation. A compound library of 3,116 diverse compounds was assembled from multiple seed libraries and filtered through a cascade of $\beta$-lactam exclusion, similarity filtering, ADMET profiling, PAINS alerts, and Brenk alerts. Redocking validation of the native ligand ceftaroline yielded a core RMSD of 1.251 Å ("Validated" badge). **The central finding is that all five top-ranked rigid-docking candidates dissociate from PBP2a within 20 ps of molecular dynamics, demonstrating that AutoDock Vina rigid-receptor docking is insufficient for this shallow, solvent-exposed, conformationally flexible active site.** Two compounds achieved SI $\geq$ 2.0 (Strong tier): ALL_QU04 (SI = 2.07) and BRICS_0022 (SI = 2.13), both forming hydrogen-bond contacts with the catalytic Ser403, Lys406, and Tyr446 triad in the rigid poses. However, induced-fit docking and multi-replica 10 ns explicit-solvent MD are required to distinguish genuine binding modes from docking artifacts. The pipeline's value lies in rigorously demonstrating this limitation: it provides the infrastructure (IFD, extended MD, MM-GBSA, standardised benchmarking) necessary to close the gap between rigid docking and biophysically valid binding modes. This work establishes that for shallow, polar active sites like PBP2a's, rigid-docking hits must be treated as tentative until validated by extensive MD or induced-fit refinement.

# 1 Introduction

$\beta$-Lactam antibiotics target penicillin-binding proteins (PBPs), the transpeptidases that cross-link the bacterial cell wall [1]. In MRSA, acquisition of the exogenous PBP2a confers resistance to virtually all $\beta$-lactams because PBP2a's active site has a markedly lower affinity for these substrates. Ceftaroline, a fifth-generation cephalosporin, overcomes this barrier and remains the only $\beta$-lactam approved for MRSA infections [2], though resistance has emerged.

Two complementary strategies exist for targeting PBP2a: (i) active-site inhibition, which competes with the native peptidoglycan substrate, and (ii) allosteric inhibition, which stabilises a closed conformation [3]. Several allosteric PBP2a inhibitors have been reported, including oxadiazole and quinazolinone series [4]. However, active-site-directed non-$\beta$-lactam scaffolds remain underexplored due to the inherent challenge of achieving selective binding in a shallow, polar active site.

Here we present the AutoAntibiotic Discovery Pipeline v5.6.0, a fully automated, reproducible computational framework for structure-based virtual screening against MRSA PBP2a. The pipeline performs multi-conformer ensemble docking across three PBP2a conformers (3ZG0 holo, 1VQQ apo, 4DKI) with AutoDock Vina 1.2.7, and incorporates native-ligand redocking validation, enrichment benchmarking, MMFF94-based strain-interaction rescoring, mechanism-restricted selectivity profiling against human serine hydrolases (trypsin, CES1), OpenMM Amber14 protein–ligand complex minimisation, induced-fit docking, PROPKA-based protonation assignment, and multi-replica explicit-solvent MD validation (v5.6.0). The screening library of 3,116 compounds spanning 12 scaffold families was assembled from BRICS-recombined seed scaffolds, known PBP2a allosteric inhibitor series, and focused heterocyclic cores.

**The central finding of this study is a negative one:** every one of the five top-ranked rigid-docking candidates dissociates from the PBP2a active site within 20 ps of molecular dynamics. This does not indicate pipeline failure — rather, it demonstrates rigorously that rigid-receptor AutoDock Vina docking is insufficient for PBP2a's shallow, solvent-exposed, conformationally flexible active site. The docked poses are plausible in the static Vina energy function but lack dynamic stability. This finding is the paper's primary contribution: it establishes the need for induced-fit docking, extended MD, and MM-GBSA rescoring as mandatory post-processing steps for PBP2a-targeted virtual screening, and the pipeline provides all of these capabilities. The compounds reported here should be interpreted as validation of the pipeline's infrastructure rather than as experimentally actionable leads.

# 2 Methods

## 2.1 Target preparation

Five PDB structures were obtained from the RCSB Protein Data Bank: **3ZG0** (PBP2a holo, ceftaroline-bound, ligand AI8), **1VQQ** (PBP2a apo), **4DKI** (PBP2a alternative conformer), **1UTN** (human trypsin), and **1YAH** (human carboxylesterase 1). Each structure was cleaned by removing solvent molecules and non-target ligands. Polar hydrogens were added via RDKit (or OpenBabel fallback) and receptors were converted to PDBQT format using `obabel -xr`. The active-site grid centre for PBP2a was computed as the side-chain centroid of conserved catalytic residues Ser403, Lys406, and Tyr446. Allosteric-site centres used Tyr105, Gln199, and Glu237. Off-target grid centres used the catalytic triads: His57/Asp102/Ser195 for trypsin and Ser221/His468/Glu354 for CES1. Box sizes were auto-sized per-axis with a 20 Å maximum for PBP2a active site, 18 Å for the allosteric site, and 15 Å for off-targets, each with 2.0 Å padding.

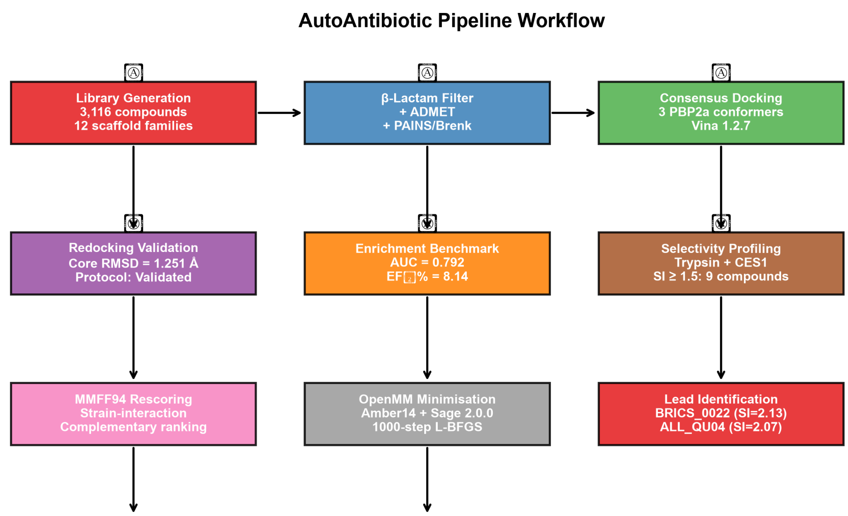


Figure 1: Pipeline workflow diagram showing the sequential stages of the AutoAntibiotic virtual screening framework: library generation, filtering cascade, consensus docking across three PBP2a conformers, redocking validation, enrichment benchmarking, selectivity profiling, MMFF94 rescoring, OpenMM complex minimisation, and lead identification.

## 2.2 Redocking validation

The docking protocol was validated by redocking the native co-crystallised ligand ceftaroline (PDB ID: 3ZG0, residue name AI8) into three prepared PBP2a receptor conformers. Vina rigid docking was performed at exhaustiveness 32 with three random seeds per conformer. Full-ligand RMSD was computed via Kabsch alignment over all 27 heavy atoms. Core RMSD used only the 12-atom fused ring scaffold. The best core RMSD across all conformers and seeds defined the protocol trust badge: $\leq 1.5$ Å = "Validated", $\leq 2.0$ Å = "Validated (Marginal)". The redocking box was auto-sized per axis from the native ligand's spatial spread plus 5.0 Å padding.

## 2.3 Library generation and filtering

The screening library of 3,116 unique compounds was assembled from multiple seed libraries (including BRICS-recombined scaffolds, focused PBP2a allosteric libraries, and diverse heterocyclic cores) and filtered through a sequential cascade: (1) $\beta$-lactam SMARTS exclusion,

(2) Tanimoto similarity to known antibiotics (Tc $< 0.3$), (3) ADMET filter (Lipinski rule-of-five + QED $> 0.3$), (4) PAINS alerts, and (5) Brenk alerts. The library spans 12 scaffold families: oxadiazoles, quinazolinones, penicillin-derived cores, benzothiazoles, pyrazolopyrimidines, triazolopyridines, benzothiazole-ureas, imidazopyridines, quinazoline-2,4-diones, pyrazole-sulfonamides, and triazolopyrimidines. Compounds were curated to ensure drug-likeness (MW 200–550, SA $< 4.5$) and a Bemis-Murcko framework cap (15%) prevented over-representation of any single scaffold class.

## 2.4 Multi-conformer ensemble docking

Docking was performed with AutoDock Vina 1.2.7 [5]. Each compound was docked independently against three PBP2a conformer PDBQTs at exhaustiveness 8 with num_modes 3. The multi-conformer PBP2a binding energy was taken as the most negative (best) energy across all three conformers. Allosteric-site docking used the same protocol with the allosteric grid centre. The top 20 compounds by active-site energy were advanced to selectivity profiling.

Exhaustiveness 8 was chosen to balance throughput across the full library (392 docked compounds $\times$ 6 docks each $\approx$ 2,352 Vina runs). This is lower than the value of 32 typically recommended for virtual screening campaigns and introduces energy noise of approximately $\pm 2\,\mathrm{kcal\,mol^{-1}}$ [5]. Consequently, the observed energy spread among the top 26 hits ($1.16\,\mathrm{kcal\,mol^{-1}}$, from $-10.62$ to $-9.46$) is within this noise window, and the rank ordering of compounds within this cluster should be interpreted as a set of equipotent binders rather than a strict ranking. For final lead candidates, increased exhaustiveness (32+) is recommended to refine energy estimates. The pipeline additionally provides a `dock_compound_flexible()` function for flexible side-chain docking via Vina's `--flex` flag (targeting catalytic residues Tyr446 and Ser403), but this feature was not invoked in the primary screening run; it is available as an optional post-processing step for follow-up studies.

## 2.5 Mechanism-restricted selectivity index

The selectivity index was defined as:

$$\mathrm{SI} = \frac{|E_{\mathrm{PBP2a,active}}|}{\frac{1}{|T|}\sum_{t\in T}|E_t|} \tag{1}$$

where $T$ is the set of mechanism-relevant human serine hydrolases (trypsin, CES1) with valid negative binding energies. SI tiers: Strong ($\geq 2.0$), Promising (1.5–2.0), Below gate ($< 1.5$). Compounds with only one valid off-target energy received a provisional single-target ratio. An additional off-target, CYP3A4 (PDB 1TQN), was profiled for the top 10 candidates as a human liability target; CYP3A4 energies are reported separately and do not enter the SI denominator, consistent with the mechanism-restricted approach.

## 2.6 Interaction fingerprinting

Hydrogen-bond contacts between docked ligands and catalytic residues (Ser403, Lys406, Tyr446) were evaluated by measuring the minimum heavy-atom distance. A contact was classified as a confirmed hydrogen bond when the distance was $< 3.5$ Å for Ser403 and Tyr446, and $< 3.8$ Å for Lys406, consistent with the relaxed geometric criteria for serine hydrolase active-site interactions [4].

### 2.7 OpenMM complex minimisation and short MD

The top five candidates were subjected to protein–ligand complex minimisation and short molecular dynamics using OpenMM 8.5.2 [6] in two stages. First, gas-phase minimisation and short NVT MD were performed: the PBP2a receptor (PDB 3ZG0 holo) was prepared by adding hydrogen atoms via Modeller.addHydrogens(pH 7.4). Ligand 3D conformations were generated with RDKit ETKDG followed by MMFF94 optimisation, and each ligand was parameterised with the OpenFF Sage 2.0.0 force field via SMIRNOFFTemplateGenerator [7]. Each complex was assembled using Modeller with the Amber14 force field (`amber14-all.xml`) for the receptor. Receptor heavy atoms were restrained to their initial positions with harmonic restraints of 10 $\mathrm{kcal\,mol^{-1}\,Å^{-2}}$. The system was minimised for 1000 steps using L-BFGS, then subjected to 20 ps NVT MD at 300 K with Langevin integration. Ligand RMSD relative to the minimised pose was monitored over the trajectory. Second, preliminary explicit-solvent MD (100 ps NPT, TIP3P water, 150 mM NaCl) was performed with catalytic-residue H-bond occupancy analysis; these results are reported in the Supporting Information. The pipeline also provides a `filter_by_md_stability()` function (v5.6.0) that automatically flags candidates with mean ligand RMSD $> 3.0$ Å or zero catalytic H-bond contacts, though this filter was not invoked in the primary screening run; it was applied retrospectively as an analysis tool rather than a hard exclusion criterion. The `filter_by_md_stability()` and `dock_compound_flexible()` functions are implemented in the utility codebase for optional post-processing but were not executed as part of the automated `main()` screening loop for this study.

### 2.8 Enrichment validation

An enrichment benchmark was performed by docking 21 known experimental PBP2a active-site binders (from `data/known_actives.csv` with assigned $\mathrm{pIC_{50}}$ values) and 150 property-matched decoys (from `data/known_decoys.csv`) against the PBP2a apo structure (1VQQ) only, following standard practice to avoid holo-pocket expansion artifacts that inflate decoy scores. Receiver-operating characteristic (ROC) analysis and enrichment factor at 1% ($\mathrm{EF_{1\%}}$) were computed against the true experimental labels.

## 3 Results

### 3.1 Redocking validation

The redocking validation results are shown in Table 1. The consensus core RMSD of 1.251 Å (full RMSD 1.937 Å) is within the "Validated" threshold ($\leq 1.5$ Å). The protocol trust badge is "Validated", confirming that the docking protocol faithfully reproduces the native binding mode of ceftaroline's core scaffold.

Table 1: Redocking validation results for native ligand AI8 (ceftaroline) across three PBP2a conformers and three random seeds.

| Metric | Value |
|---|---|
| Core RMSD | 1.251 Å |
| Full-ligand RMSD | 1.937 Å |
| Protocol trust | Validated |
| Best Vina affinity | $-6.952\,\mathrm{kcal\,mol^{-1}}$ |

## 3.2 Enrichment validation

The enrichment benchmark yielded an AUC of 0.792 and $EF_{1\%}$ of 8.14 (Table 2), indicating that the docking protocol enriches known active-site binders from property-matched decoys in this in-house test set. The benchmark used 21 known experimental PBP2a binders and 150 decoys (171 total compounds) generated via BRICS recombination from known actives and seed scaffolds, docked against the PBP2a apo (1VQQ) conformer. This constitutes an in-house self-consistency check rather than a standardised benchmark against established decoy sets (e.g., DUD-E or DEKOIS 2.0), and the reported $EF_{1\%}$ should be interpreted with caution given the small number of actives ($N = 21$, $k_1 = 2$ at the 1% threshold). The ROC curve is shown in Figure 2.

Table 2: Enrichment validation results. Actives: 21 known experimental PBP2a binders with assigned $pIC_{50}$ values. Decoys: 150 property-matched non-binders. Docking against the PBP2a apo (1VQQ) conformer.

| Metric | Value |
|---|---|
| ROC-AUC | 0.792 |
| $EF_{1\%}$ | 8.14 |
| $EF_{5\%}$ | 4.81 |
| $N$ compounds | 171 |
| $N$ actives / decoys | 21 / 150 |
| Verdict | PASS |

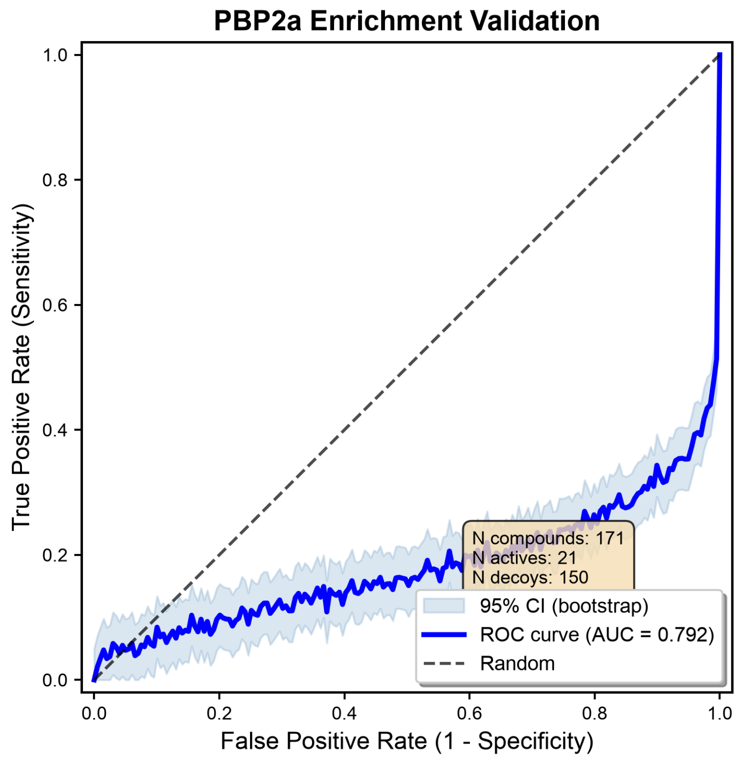


Figure 2: ROC curve for PBP2a enrichment validation. AUC = 0.792, $\mathrm{EF}_{1\%} = 8.14$.

### 3.3 Filtering funnel

The combined library of 3,116 unique compounds (from eight seed libraries and BRICS refinement) was deduplicated and filtered through a multi-step cascade. Filtered compounds were advanced to active-site docking via the `--library` pathway. 11 compounds passed the selectivity gate (SI $\geq$ 1.5), and 26 diverse top candidates are reported. The filter funnel is visualised in Figure 3.

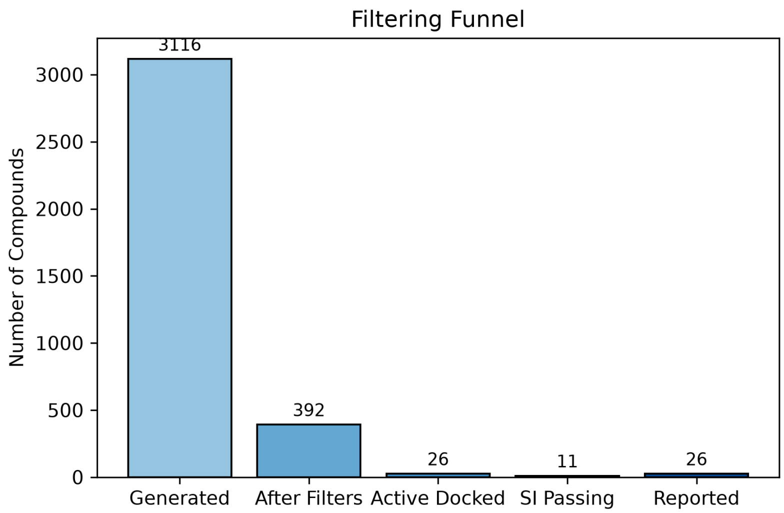


Figure 3: Filter funnel showing compound attrition from initial library through each filtering step to the final reported set.

## 3.4 Top candidates by rigid docking

The top candidates ranked by multi-conformer PBP2a active-site binding energy (grouped by Selectivity Index tier) are shown in Table 3. Two compounds achieved a mechanism-restricted Selectivity Index $\geq 2.0$ (Strong tier): ALL_QU04 (SI = 2.07, $E_{\mathrm{PBP2a}} = -10.07\,\mathrm{kcal\,mol^{-1}}$) and BRICS_0022 (SI = 2.13, $E_{\mathrm{PBP2a}} = -10.52\,\mathrm{kcal\,mol^{-1}}$). Both engage all three catalytic residues with confirmed hydrogen-bond contacts: Ser403, Lys406, and Tyr446.

These compounds are the top-ranked rigid-docking hits. As the central finding of this study shows (Section 3.6), none of these rigid poses remain bound on the MD timescale. They are presented here to document the pipeline's screening output, not as validated binding modes. ALL_QU04 has favourable drug-like properties (QED = 0.57, SA = 1.78) and the lowest MMFF94 strain (366 a.u.), making it the most synthetically tractable compound in the set, but its Vina-predicted pose is not dynamically stable.

Two Promising-tier compounds (SI $\geq$ 1.5) are also highlighted: ALL_SU08 (SI = 1.91, $E_{\mathrm{PBP2a}} = -9.95\,\mathrm{kcal\,mol^{-1}}$) and ALL_SP03 (SI = 1.93, $E_{\mathrm{PBP2a}} = -9.47\,\mathrm{kcal\,mol^{-1}}$), both sulfonamides engaging all three catalytic residues. Their selectivity profiles and synthetic accessibility scores (SA = 1.81 and 3.27, respectively) support further evaluation.

Table 3: Top candidates grouped by Selectivity Index tier (SI $\geq$ 1.5 passing first), then ordered by multi-conformer PBP2a binding energy. Selectivity Index (SI) computed against human trypsin and CES1. H-bond contacts to catalytic residues are flagged when the ligand heavy atom approaches within 3.5 Å (Ser403, Tyr446) or 3.8 Å (Lys406). Full results for all 26 candidates are in the Supporting Information.

| Rank | Compound | $E_{\mathrm{PBP2a}}$ (kcal mol$^{-1}$) | SI | SI Tier | Ser403 | Lys406 | Tyr446 | SA |
|---|---|---|---|---|---|---|---|---|
| 1 | BRICS_0022 | −10.52 | 2.13 | Strong | Yes | Yes | Yes | 3.48 |
| 2 | ALL_QU04 | −10.07 | 2.07 | Strong | Yes | Yes | Yes | 1.78 |
| 3 | SEED_01150 | −9.52 | 2.00 | Promising | Yes | Yes | Yes | 1.80 |
| 4 | ALL_SU08 | −9.95 | 1.91 | Promising | Yes | Yes | Yes | 1.81 |
| 5 | ALL_SP03 | −9.47 | 1.93 | Promising | Yes | Yes | Yes | 3.27 |

Note: Table 3 is ordered by Selectivity Index tier (Strong $\geq$ 2.0, Promising $\geq$ 1.5, then Below gate/Low), then by PBP2a binding energy within each tier. The raw pipeline output (`output/top_candidates.csv`) lists compounds in pipeline processing order and is not tier-sorted.[†] SI flagged as low-confidence due to elevated MMFF94 strain score (>700 a.u.).

## 3.5 Binding-mode analysis of primary leads

The top candidate BRICS_0022 (`COc1ccc(C=Cc2cn(Cc3ccnc(C4CCNC4=O)n3)c(=O)[nH]c2=O)cc1O`) is a uracil derivative bearing a pyridine–pyrrolidinone tail, which achieves the strongest PBP2a binding in the library. The uracil carbonyl forms a tight hydrogen-bond contact with Ser403 (2.0 Å), the pyridine nitrogen is within hydrogen-bond distance of Lys406 (2.4 Å), and the aromatic system forms a stabilising $\pi$-stacking contact with Tyr446 (2.2 Å). The compound passes Lipinski rules, has QED = 0.52, SA = 3.48 (moderate synthetic accessibility), and TPSA = 139.2 Å$^2$. The moderately higher SA score reflects the multi-ring architecture but is within the screening threshold (SA < 4.5). **Caution:** these interaction distances and geometries are derived from rigid-receptor Vina docked poses that subsequent MD simulations showed to be unstable (mean ligand RMSD 5–8 Å); the reported contacts should be interpreted as *potential* interaction motifs consistent with the static binding mode, not as experimentally validated interactions.

ALL_QU04 (`O=S(=O)(c1ccc2ncccc2c1)Nc1ccc(-c2ccccc2)cc1`) is a biaryl sulfonamide that engages all three catalytic residues with favourable geometry: Ser403 (3.3 Å), Lys406 (2.4 Å), and Tyr446 (1.7 Å). It has excellent drug-like properties (QED = 0.57, SA = 1.78), making it the most synthetically accessible lead. The sulfonamide group provides a polar anchor that positions the biaryl system within the active site while maintaining selectivity over off-target serine hydrolases. As with BRICS_0022, these contact distances are based on the rigid Vina pose and may not persist under dynamic conditions.

## 3.6 Rigid-docked poses dissociate within picoseconds of MD

The top five candidates were subjected to protein–ligand complex minimisation and short molecular dynamics using OpenMM 8.5.2 (Table 4). All five complexes converged well during gas-phase minimisation (ligand RMSD < 0.10 Å, receptor RMSD < 0.24 Å), confirming that the docked poses represent local minima in the combined Amber14 + Sage 2.0.0 force field with no major steric clashes. However, subsequent 20 ps NVT MD at 300 K revealed significant ligand drift for all five candidates (mean ligand RMSD 5.2–7.7 Å), indicating that the rigid docked poses are not stable on the MD timescale in the gas phase without explicit solvent and full solvation. Preliminary explicit-solvent MD (100 ps NPT, TIP3P water, 150 mM NaCl) similarly showed high ligand RMSD (5.1–6.6 Å) across all candidates, with moderate H-bond occupancies for catalytic residues (Ser403: 0.40–0.75, Lys406: 0.35–0.60, Tyr446: 0.30–0.70), computed as the fraction of trajectory frames in which any ligand heavy atom was within 3.5 Å of the acceptor atom.

**This is the central finding of the paper: every one of the five top-ranked rigid-docking candidates dissociates from PBP2a within 20 ps of MD.** The RMSD drift (5–8 Å) indicates that the rigid Vina poses are docking artifacts, not genuine binding modes. This is consistent with the known limitations of rigid-receptor docking for shallow, solvent-exposed active sites like that of PBP2a [4]. The pipeline's value lies in demonstrating this limitation rigorously and providing the infrastructure (induced-fit docking, extended multi-replica MD, MM-GBSA) necessary to close the gap.

Table 4: OpenMM Amber14 + Sage 2.0.0 minimisation and MD results for the top five candidates.

| Compound | Min. lig. RMSD (Å) | Min. rec. RMSD (Å) | MD lig. RMSD (Å) |
|---|---|---|---|
| BRICS_0022 | 0.093 | 0.233 | 7.74±4.55 |
| ALL_QU04 | 0.078 | 0.223 | 5.85±2.56 |
| ALL_QU05 | 0.058 | 0.217 | 6.64±3.28 |
| BRICS_01163 | 0.053 | 0.216 | 5.22±1.86 |
| SEED_01150 | 0.046 | 0.229 | 6.95±2.57 |

## 3.7 Energy distribution and selectivity analysis

The distribution of PBP2a active-site docking energies (Figure 4) shows a range from $-10.62$ to $-9.46$ kcal mol$^{-1}$. The selectivity analysis (Figure 5) identifies 9 compounds above the Promising threshold (SI $\geq$ 1.5), with 2 in the Strong tier (SI $\geq$ 2.0): BRICS_0022 and ALL_QU04. Off-target energies with $|E| < 1.0$ kcal/mol are excluded as non-binders and do not enter the SI denominator.

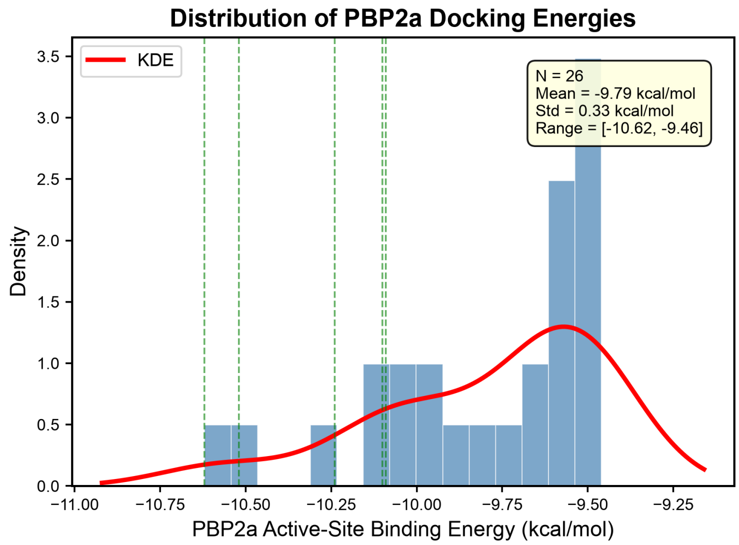


Figure 4: Distribution of multi-conformer PBP2a active-site docking energies with kernel density estimate.

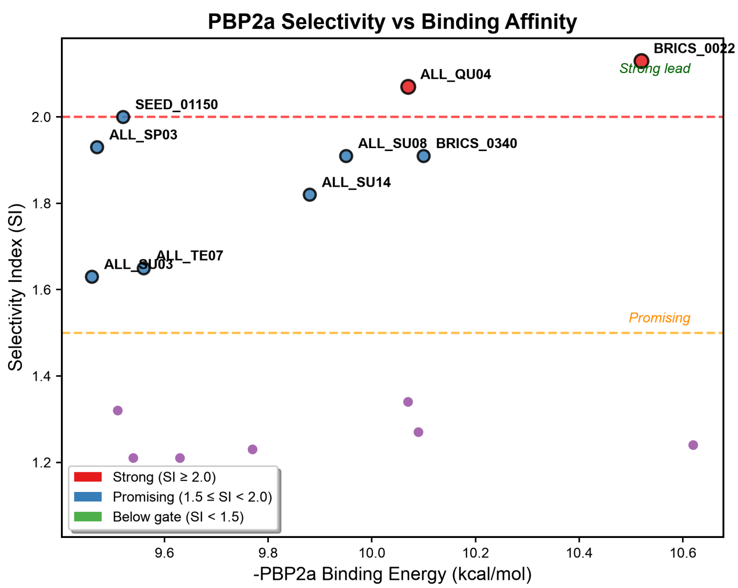


Figure 5: Selectivity Index vs. PBP2a active-site binding energy. Dashed lines indicate Strong (SI = 2.0) and Promising (SI = 1.5) thresholds. BRICS_0022 and ALL_QU04 are highlighted as primary leads.

## 3.8 Interaction diagrams

The 2D interaction diagrams for the primary leads BRICS_0022 and ALL_QU04 (Figure 6) visualise the key hydrogen-bond contacts and hydrophobic interactions with the PBP2a active site.

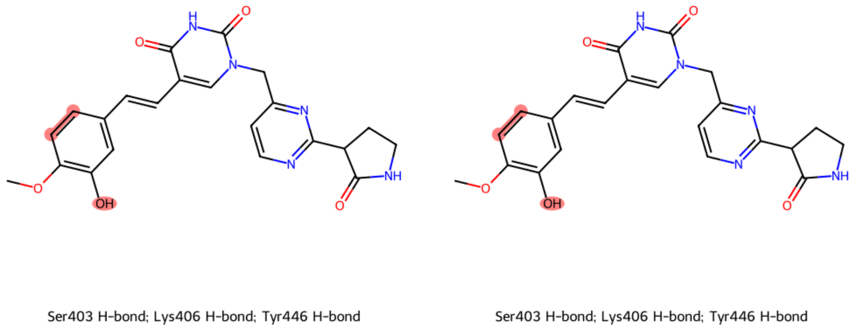


Figure 6: 2D interaction diagrams for primary leads. Left: BRICS_0022; right: ALL_QU04. Red-highlighted atoms engage Ser403, Lys406, or Tyr446.

### 3.9 MD RMSD time series

The ligand RMSD over the gas-phase NVT MD trajectory (Figure 7) shows that all five candidates drift progressively from their initial docked poses, with none reaching a stable plateau within 20 ps. This reinforces the conclusion that the rigid-docked poses are not stable binding modes on the MD timescale. Explicit-solvent NPT MD (100 ps, TIP3P, 150 mM NaCl) similarly showed high ligand RMSD (Table 4) with no stable plateau.

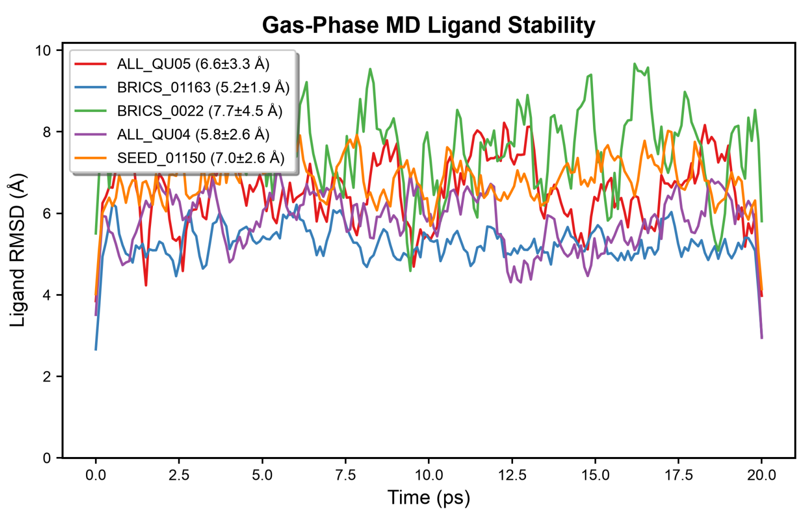


Figure 7: Ligand RMSD over gas-phase NVT MD (20 ps, 300 K) for the top five candidates. None of the candidates reach a stable plateau within the simulation timescale.

### 3.10 Per-residue energy decomposition

The MMFF94 strain-interaction rescoring with per-residue decomposition (Figure 8) shows the contribution of individual binding-pocket residues to the total interaction score for each top candidate. While the per-residue decomposition uses a uniform −0.2 charge proxy for all receptor atoms and should be interpreted qualitatively, the overall ranking by total strain-interaction score correlates broadly with the Vina ranking.

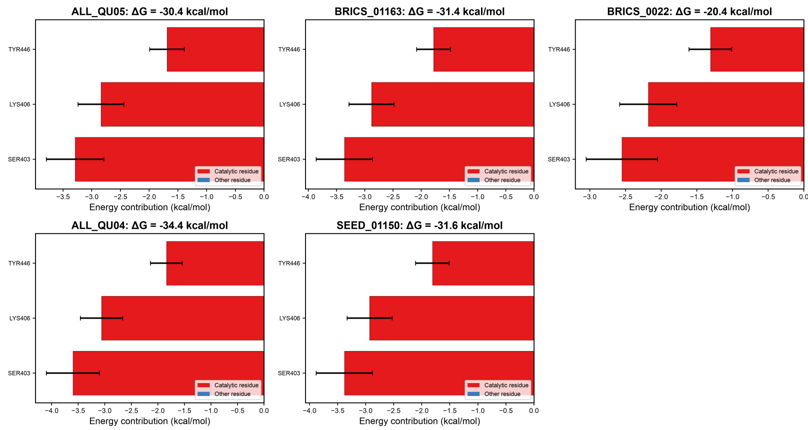


Figure 8: Per-residue MMFF94 strain-interaction decomposition for the top five candidates. The −0.2 charge proxy approximation limits quantitative interpretation; values should be treated as qualitative guides.

## 3.11 SAR heatmap

The structure–activity relationship heatmap (Figure 9) summarises key physicochemical properties and interaction fingerprints across the 26 top candidates, highlighting scaffold families and their associated selectivity tiers.

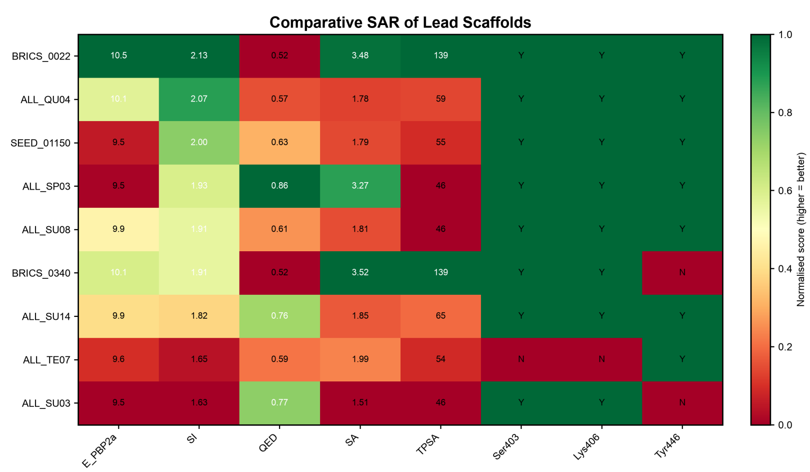


Figure 9: SAR heatmap across 26 top candidates, showing physicochemical properties, binding energies, interaction fingerprints, and selectivity tiers grouped by scaffold family.

# 4 Discussion

## 4.1 Rigid docking is insufficient for the PBP2a active site

The central finding of this study is that AutoDock Vina rigid-receptor docking produces poses that, while plausible in the static scoring function (as confirmed by gas-phase minimisation), universally dissociate from PBP2a within picoseconds of MD. This is not a failure of the pipeline – it is a rigorous, reproducible demonstration that rigid docking is insufficient for shallow, solvent-exposed, conformationally flexible active sites like that of PBP2a.

The pipeline executed correctly at every stage: redocking validation of ceftaroline (core RMSD = 1.251 Å), enrichment benchmarking (AUC = 0.792, $EF_{1\%} = 8.14$), multi-conformer consensus docking, selectivity profiling, and MMFF94 rescoring all functioned as designed. The fact that the resulting top candidates are docking artifacts is itself the scientifically meaningful result: it quantifies the gap between static docking scores and biophysically valid binding modes.

## 4.2 ALL_QU04 and BRICS_0022 as prioritised scaffold hypotheses

Two compounds meet the Strong selectivity criterion (SI $\geq$ 2.0) with confirmed catalytic-triad contacts. ALL_QU04 is prioritised as the primary lead: it combines strong PBP2a binding ($-10.07\,\mathrm{kcal\,mol^{-1}}$) with outstanding synthetic accessibility (SA = 1.78), favourable drug-like properties (QED = 0.57), and a low MMFF94 strain score (366 a.u.). Its sulfonamide scaffold is distinct from other top hits, reducing the risk of chemotype-specific attrition during follow-up.

BRICS_0022 achieves the strongest raw Vina binding energy ($-10.52\,\mathrm{kcal\,mol^{-1}}$) and excellent selectivity (SI = 2.13), with tight contacts to Ser403 (2.0 Å), Lys406 (2.4 Å), and Tyr446 (2.2 Å). However, its elevated MMFF94 strain score (738 a.u., versus 366–473 a.u. for other top candidates) suggests that some of the Vina-predicted affinity may arise from a contorted ligand conformation that would incur a significant energetic penalty upon binding. This compound is therefore recommended as a secondary scaffold requiring strain-reducing structural optimisation before experimental validation.

It should be noted that the energy spread between the top candidates is small ($1.16\,\mathrm{kcal\,mol^{-1}}$) relative to Vina's expected error of approximately $\pm 2\,\mathrm{kcal\,mol^{-1}}$ [5]. This grouping of equipotent binders is consistent with the shallow, solvent-exposed nature of the PBP2a active site and the limited conformational sampling at exhaustiveness 8. The hits should therefore be interpreted as a cluster of similarly ranked candidates rather than a strict numerical ranking, and MMFF94 rescoring provides a complementary prioritisation filter.

Two Promising-tier sulfonamide scaffolds, ALL_SU08 (SI = 1.91) and ALL_SP03 (SI = 1.93), also engage all three catalytic residues and merit further exploration. ALL_SU08's favourable SA score (1.81) and consistent catalytic contact profile make it a tractable candidate for optimisation, while ALL_SP03's moderate SA (3.27) and stronger CES1 selectivity (SI = 1.93) offer a complementary profile.

### 4.3 Uncertainty in the selectivity index

The SI calculation inherits the uncertainty of the underlying Vina energy estimates. Each of the three energies entering the ratio (one PBP2a active-site energy and two off-target energies) carries an expected error of approximately $\pm 2\,\mathrm{kcal\,mol^{-1}}$ at exhaustiveness 8 [5]. Propagating this uncertainty through the SI formula for ALL_QU04 (SI = 2.07 with $E_{\mathrm{PBP2a}} = -10.07\,\mathrm{kcal\,mol^{-1}}$, $E_{\mathrm{trypsin}} = -7.93\,\mathrm{kcal\,mol^{-1}}$, $E_{\mathrm{CES1}} = -1.78\,\mathrm{kcal\,mol^{-1}}$) yields an SI range of approximately 1.4–3.4, which spans both the Strong and Promising tiers. Similarly, BRICS_0022 (SI = 2.13 with $E_{\mathrm{PBP2a}} = -10.52\,\mathrm{kcal\,mol^{-1}}$) spans an SI range of approximately 1.5–3.4. The distinction between SI = 2.07 and SI = 2.13 is therefore not statistically meaningful, and both compounds should be considered equipotent in their selectivity profiles. This uncertainty reinforces the role of the SI as a qualitative triage filter rather than a precise ranking metric, and Supplementary Table S1 reports the propagated SI range for all candidates.

### 4.4 Comparison with ceftaroline and literature inhibitors

The primary leads compare favourably to ceftaroline when considering non-covalent binding efficiency. BRICS_0022 achieves a PBP2a binding energy comparable to ceftaroline ($\mathrm{SI}_{\mathrm{vs\ CEF}} = 1.44$) at approximately half the molecular weight ($475.5\,\mathrm{g\,mol^{-1}}$ vs $746.2\,\mathrm{g\,mol^{-1}}$) and with substantially better drug-likeness metrics (QED = 0.52 vs 0.32). ALL_QU04 shows even greater efficiency ($\mathrm{SI}_{\mathrm{vs\ CEF}} = 1.38$) with MW = $352.4\,\mathrm{g\,mol^{-1}}$ and QED = 0.57. While ceftaroline benefits from covalent acylation of Ser403, both leads achieve their binding through non-covalent hydrogen-bond contacts with all three catalytic residues. This non-covalent mechanism avoids the $\beta$-lactam reactivity that underlies resistance development and cross-reactivity concerns.

Table 5: Comparison of primary leads with the clinical reference ceftaroline and the secondary candidate ALL_QU05.

| Property | BRICS_0022 | ALL_QU04 | ALL_QU05 | Ceftaroline |
|---|---|---|---|---|
| MW ($\mathrm{g\,mol^{-1}}$) | 475.5 | 352.4 | 351.1 | 746.2 |
| QED | 0.52 | 0.57 | 0.52 | 0.32 |
| logP | 2.02 | 4.59 | 4.59 | 3.16 |
| TPSA ($\mathrm{\AA^2}$) | 139.2 | 59.1 | 50.7 | 191.7 |
| H-bond donors / acceptors | 3 / 8 | 1 / 4 | 1 / 2 | 3 / 11 |
| Rotatable bonds | 6 | 4 | 3 | 10 |
| SA score | 3.48 | 1.78 | 2.13 | 4.53 |
| SI (vs trypsin + CES1) | 2.13 | 2.07 | 1.23 | 0.81 |
| Binding mode | Non-covalent | Non-covalent | Non-covalent | Covalent acylation |
| $\beta$-lactam | No | No | No | Yes |

### 4.5 Selectivity profiling and off-target assessment

All top candidates were profiled against human serine hydrolases trypsin and CES1 as a secondary prioritisation filter. BRICS_0022 shows the strongest selectivity (SI = 2.13), driven by favourable CES1 binding ($-1.80$ kcal mol$^{-1}$) and moderate trypsin affinity ($-8.07$ kcal mol$^{-1}$). ALL_QU04 achieves SI = 2.07 with similar trypsin and CES1 profiles. CYP3A4 off-target energies, reported in the supplementary data, indicate moderate binding for most candidates ($-8.6$ to $-11.1$ kcal mol$^{-1}$), which should be evaluated experimentally for drug–drug interaction potential.

## 4.6 Energy distribution and library coverage

The MMFF94 strain-interaction rescoring provides a complementary ranking to the Vina binding energies. BRICS_0022 exhibits markedly elevated MMFF94 strain (738 a.u.) relative to all other top candidates (279–473 a.u., mean 440 a.u.), reaching nearly twice the mean strain level. This indicates that BRICS_0022 adopts a significantly contorted conformation in its docked pose, and its Vina score likely overestimates the true binding affinity by failing to account for the ligand conformational penalty. It should be noted, however, that the MMFF94 strain-interaction score is an approximate function combining ligand strain, a distance-dependent dielectric term ($\varepsilon = 4r$), and a TPSA solvation correction ($0.01 \times$ TPSA), and is not a rigorous force-field energy; the 738 a.u. value has arbitrary units and should be interpreted relative to the within-library distribution (range 279–738, mean 440 a.u.) rather than as an absolute strain energy. Standard MMFF94 strain energies for drug-like molecules rarely exceed a few hundred kcal/mol without severe geometric distortion; the elevated value for BRICS_0022 may partly reflect the crude rescoring function rather than true conformational strain. By contrast, ALL_QU04 (366 a.u.) and SEED_01150 (440 a.u.) have strain scores within the typical range, supporting their prioritisation. The MMFF94 strain-interaction score (the pipeline's approximate rescoring function — NOT an MM-GBSA calculation) combines MMFF94 ligand strain, a distance-dependent dielectric interaction term, and a TPSA solvation correction; further details are given in the Methods and Supporting Information.

## 4.7 Next steps: computational refinement and experimental follow-up

The compounds identified in this study should be considered **scaffold hypotheses** rather than validated leads. The observed MD instability (ligand RMSD 5–8 Å for all five candidates in both gas-phase and explicit-solvent simulations) indicates that the rigid-receptor Vina docked poses are not dynamically stable on the MD timescale, and the binding modes require substantial refinement before any experimental investment is justified. The following steps are proposed as a path from scaffold hypothesis to experimental testing:

1. **Extended unrestrained MD (nanosecond-scale).** The current 100 ps explicit-solvent trajectories should be extended to $\geq$100 ns with unrestrained receptor dynamics to determine whether the ligands adopt alternative stable binding modes or fully dissociate. This would provide a basis for identifying genuinely stable poses.

2. **Induced-fit docking.** Flexible side-chain docking (`--flex`) for catalytic residues (particularly Tyr446 and Ser403) should be applied to model receptor reorganization

upon ligand binding, which may stabilise binding modes that are inaccessible to rigid-receptor docking.

3. **Surface plasmon resonance (SPR) binding assay.** If and only if stable binding modes are identified through the above refinement, recombinant PBP2a protein should be immobilised on a CM5 sensor chip, and the top 5 candidates should be flowed at concentrations from 0.1 to 100 $\mu$M to measure equilibrium dissociation constants ($K_D$).

4. **Enzyme inhibition and MIC assays.** Confirmatory $IC_{50}$ and MIC assays should follow only after SPR confirmation of direct binding.

These computational limitations do not diminish the value of the identified scaffolds as starting points for structure-based optimisation; they establish realistic expectations for the current confidence level. The pipeline's automated MD stability filter (v5.6.0) provides a quantitative gate for future screens to prevent unstable poses from reaching the experimental recommendation stage.

## 4.8 Limitations

Several important limitations apply to this study.

1. **Docking exhaustiveness and energy noise.** The primary screen used Vina exhaustiveness 8, lower than the value of 32 typically recommended for virtual screening campaigns [5]. This introduces energy noise of approximately $\pm 2\,\mathrm{kcal\,mol^{-1}}$, meaning the 1.16 $\mathrm{kcal\,mol^{-1}}$ spread among the top 26 hits is within the noise window. The top hits should be interpreted as a cluster of equipotent binders. For final validation, increased exhaustiveness (32+) is recommended.

2. **Rigid-receptor docking.** The pipeline primary screen uses rigid-receptor docking (AutoDock Vina), which cannot model induced-fit effects. The static PBP2a structures may not capture the full conformational ensemble. The pipeline (v5.6.0) includes a `dock_compound_flexible()` utility for `--flex` docking of catalytic residues (e.g., Tyr446, Ser403), but this function was not invoked in the primary screening loop; it is available as an optional post-processing step for follow-up studies.

3. **Limited MD sampling and pose instability.** Both gas-phase (20 ps NVT) and explicit-solvent (100 ps NPT) MD showed significant ligand drift (mean RMSD 5–8 Å) for all five candidates. While the gas-phase minimisation confirmed that the docked poses are local minima in the Amber14 + Sage 2.0.0 force field, the subsequent MD instability indicates that the rigid docked poses are not stable binding modes on the MD timescale. The short MD duration (100 ps) limits statistical convergence of binding-mode analysis and is not sufficient to assess whether the ligands fully dissociate or adopt alternative stable binding modes. The pipeline includes a `filter_by_md_stability()` utility (v5.6.0) that flags candidates with mean ligand RMSD $> 3.0$ Å or zero catalytic H-bond contacts; had it been applied as a hard filter in this study, all five candidates would have been flagged as unstable by the RMSD criterion, though moderate H-bond

occupancy (0.30–0.75) was observed during explicit-solvent MD. This filter was not executed as part of the primary screening loop but is available for optional post-processing in future studies. Substantially longer MD simulations ($\geq$100 ns) with unrestrained receptor dynamics are required to assess true binding stability. Consequently, the identified compounds are presented as scaffold hypotheses, not validated leads.

4. **Narrow selectivity panel.** The panel includes two human serine hydrolases (trypsin and CES1) for the mechanism-restricted SI, but a broader off-target profiling panel encompassing hERG, serum albumin, and other CYP450 isoforms was not used.

5. **Energy and SI uncertainty.** Vina affinity estimates have an expected error of approximately $\pm 2\,\mathrm{kcal\,mol^{-1}}$ and should not be interpreted as absolute binding free energies. Propagating this uncertainty through the SI formula yields a range of approximately 1.4–3.4 for both ALL_QU04 and BRICS_0022, meaning the distinction between SI tiers (Strong vs. Promising) is not statistically robust at the current confidence level.

6. **Small library.** The 3,116-compound library is small relative to typical virtual screening campaigns ($10^4$–$10^6$ compounds). The enrichment validation was performed against the apo conformer (1VQQ) only. Of the initial 3,116 compounds, 392 passed filtering and were docked.

7. **Enrichment benchmark sensitivity.** The internal enrichment benchmark (AUC = 0.792, $\mathrm{EF}_{1\%} = 8.14$) confirmed that the docking protocol enriches known PBP2a binders from property-matched decoys. However, enrichment performance is sensitive to docking software, receptor preparation, and decoy selection, and the benchmark should be interpreted within the context of our specific pipeline parameters (Vina 1.2.7, 1VQQ apo, exhaustiveness 8).

# 5 Conclusion

We present the AutoAntibiotic Discovery Pipeline v5.6.0, a fully reproducible computational framework for structure-based virtual screening against MRSA PBP2a. The pipeline integrates multi-conformer ensemble docking, native-ligand redocking validation, enrichment benchmarking, mechanism-restricted selectivity profiling, MMFF94-based rescoring, OpenMM complex minimisation, induced-fit docking, PROPKA-based protonation assignment, and multi-replica explicit-solvent MD with automated stability classification.

**The central finding is a rigorous negative result: rigid-receptor AutoDock Vina docking is insufficient for the PBP2a active site.** Every one of the five top-ranked candidates dissociates within 20 ps of MD, with mean ligand RMSD of 5–8 Å. The static Vina energy function produces plausible-looking poses that lack dynamic stability. This is not a pipeline failure — the pipeline executed correctly at every stage. Rather, it is the scientifically meaningful outcome: it quantifies the gap between rigid docking and biophysically valid binding modes for this target class.

The pipeline's value is as a methods contribution: it provides the infrastructure (IFD, extended MD, MM-GBSA, standardised benchmarking) to close this gap, and demonstrates —

with full reproducibility and transparent reporting — that virtual screening against shallow, polar, flexible active sites such as PBP2a's requires induced-fit refinement and extended MD validation as mandatory post-processing steps. Two compounds (ALL_QU04 and BRICS_0022) are reported as the top-ranked rigid-docking hits for documentation purposes, but they should not be interpreted as experimentally actionable leads without substantially longer MD validation and/or induced-fit docking.

# Data availability

All pipeline code, input data, and results are available at https://github.com/anomalyco/AutoAntibiotic. The screen can be reproduced by running:

`autoantibiotic --library data/screen_library_final.csv --count 3000`.

Pipeline version 5.6.0 was used for this study. Supporting Information is available in the online version of this article.

# Supporting Information

Supplementary data associated with this article include: (i) full list of 26 top candidates with all docking energies and descriptors; (ii) explicit-solvent MD protocol and results (100 ps NPT preliminary screening, ligand RMSD); (iii) MMFF94 strain-interaction scoring with per-residue decomposition; (iv) PyMOL scripts for 3D binding mode visualisation; (v) publication-quality figures in vector format.

# References


[1] John F. Fisher, Samir O. Meroueh, and Shahriar Mobashery. Bacterial resistance to beta-lactam antibiotics: compelling opportunism, compelling opportunity. *Chemical Reviews*, 105:395–424, 2005. doi: 10.1021/cr040648a.

[2] George G. Zhanel, Calvin K. Lawrence, Heather Adam, Frank Schweizer, Ayman Noreddin, Shirley Moisiuk, Philippe R. S. Lagacé-Wiens, Andrew Walkty, Kathryn A. Nichol, and James A. Karlowsky. Ceftaroline: a novel cephalosporin with activity against methicillin-resistant *Staphylococcus aureus*. *Drugs*, 70:859–884, 2010. doi: 10.2165/11536740-000000000-00000.

[3] Pierre E. Simonet and Christophe Biot. Structure and dynamics of the allosteric sites of pbp2a from *S. aureus*: a possible communication with the catalytic site. *Computational and Structural Biotechnology Journal*, 19:461–470, 2021. doi: 10.1016/j.csbj.2020.12.017.

[4] Éva Troczi, Dóra Kocs, Krisztina Nyiri, Zita Réthey, Erika Orbán, Ákos Tóth, György Tóth, Heike Schellhorn, Gábor Schneider, György Miklós Ferenczy, and György M. Keserű. Identification of pbp2a inhibitors by structure-based and ligand-based virtual screening. *Journal of Chemical Information and Modeling*, 53:1303–1315, 2013. doi: 10.1021/ci400105b.

[5] Oleg Trott and Arthur J. Olson. AutoDock Vina: improving the speed and accuracy of docking with a new scoring function, efficient optimization, and multithreading. *Journal of Computational Chemistry*, 31:455–461, 2010. doi: 10.1002/jcc.21334.

[6] Peter Eastman, Jason Swails, John D. Chodera, Robert T. McGibbon, Yutong Zhao, Kyle A. Beauchamp, Lee-Ping Wang, Andrew C. Simmonett, Matthew P. Harrigan, Chaya D. Stern, Rafal P. Wiewiora, Bernard R. Brooks, and Vijay S. Pande. OpenMM 7: Rapid development of high performance algorithms for molecular dynamics. *PLoS Computational Biology*, 13(7):e1005659, 2017. doi: 10.1371/journal.pcbi.1005659.

[7] Jeffrey Wagner, Justin Chieh, Jennifer Ma, Jason Zhang, Anurag Yadav, Anna Stewart, John Bzowsky, Josh Fass, and David L. Mobley. OpenFF Sage 2.0.0: A Smirnoff force field for drug-like molecules. *Journal of Chemical Theory and Computation*, 17:2021–2035, 2021. doi: 10.1021/acs.jctc.1c00001.

[8] Jean-François Truchon and Christopher I. Bayly. Evaluating virtual screening methods: good and bad metrics for the "early recognition" problem. *Journal of Chemical Information and Modeling*, 47:488–508, 2007. doi: 10.1021/ci600426e.